Dongwook Kwon

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Education

Kwangwoon University Seoul, Korea
M.S. in Computer Engineering, GPA 4.33 / 4.5 (98.1%) *Mar. 2024 – Feb. 2026*

University of Toronto Toronto, ON, Canada
Graduate Research Program, GPA 3.68 / 4.0 *Jan. 2025 – Jul. 2025*

Kwangwoon University Seoul, Korea
B.S. in Electronics and Communications Engineering, GPA 3.41 / 4.5 (88.1%) *Mar. 2019 – Feb. 2024*

Experience

AI Engineer, Intelligent Agent Technology Team Dec. 2025 – Present
Suresoft Technologies *Seongnam, Korea*

- Developing an AI validation system utilizing AI agent–based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Research Assistant Aug. 2021 – Dec. 2025
Bio-Computing & ML Lab, Kwangwoon University *Seoul, Korea*

- Researched deep-learning algorithms for anomaly detection in time-series data (ECG, cyber-physical systems).
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.
- Published in EMNLP 2025 (Oral) and ACL 2026; co-author on two granted Republic of Korea patents.

Visiting Researcher Jan. 2025 – Jul. 2025
LG Electronics Canada *Toronto, ON, Canada*

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

AI Development Project Participant, Summer Program Jul. 2022 – Aug. 2022
Qualcomm Institute, UC San Diego *San Diego, CA, USA*

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Projects

Agentic AI Evaluation Framework | *Python, LLMs, multi-agent systems* Jan. 2025 – Present

- Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

AI Intrusion Detection for APR1400 | *PyTorch, anomaly detection* Jul 2022 – Present

- Developed an AI defense system to protect nuclear power plants from cyber attacks.

Quadruped Robot with Vision SLAM | *ROS, depth camera, SLAM* Apr 2022 – Nov. 2022

- Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Smart Logistics System Using Computer Vision | *OpenCV, container tracking* Apr 2021 – Nov. 2021

- Developed a logistics detection system incorporating location tracking and stack-layer identification for containers. — Ministerial Gold Award.

Publications & Patents

R. Chang†, **D. Kwon**†, J. Lee†, and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades." *ACL 2026 — Industry Track (Poster).* †Equal contribution.

J. Lee†, R. Chang†, **D. Kwon**†, H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems." *EMNLP 2025 — Industry Track (Oral).* †Equal contribution.

D. Kwon, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid GAT + Transformer for implantable-device anomaly detection." *IBEC 2024.* **Best Poster Award — Silver.**

D. Kwon, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model." *KOSOMBE 2023.* **Outstanding Paper Award.**

D. Kwon, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments." Under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

Patent **KR 10-2961984** — Real-time anomaly detection for implantable medical devices (Apr 2026).

Patent **KR 10-2848814** — GPT API-Based Network Anomaly Detection (Aug 2025).

Technical Skills

Languages: Python, C/C++, MATLAB, Bash
Frameworks: PyTorch, TensorFlow, Transformers, LangChain, NumPy, Pandas, scikit-learn
Developer Tools: Linux, Docker, Git, CUDA, VS Code
Domains: Agentic AI, multi-agent LLM systems, GNNs, time-series anomaly detection, biomedical signals